

Definitely Not An Analysis Midterm

Time: 50 minutes

Oct. 2025

Problem 1. (56 pts)

For each of the following statements, write a specific example if possible. If not possible, simply state so. No proofs are required. All spaces are metric spaces, but not necessarily $\mathbb{R}^n$ unless otherwise stated.

1. A sequence (x_n) in $\mathbb{R}^2$ that converges in the Euclidean metric

$$d_2(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

but does not converge in the sup metric

$$d_\infty(x, y) = \max\{|x_1 - y_1|, |x_2 - y_2|\}.$$

2. A continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ that maps compact sets to compact sets but which is not open.
3. A disconnected set A such that its closure $\overline{A}$ is connected.
4. A metric space (X, d) with infinitely many points and exactly eight clopen subsets.
5. A countable set $A \subset \mathbb{R}$ whose closure $\overline{A}$ is uncountable.
6. A closed and bounded but non-compact set.

Problem 2. (44 pts)

Consider the set

$$S = \left\{ (x_i)_{i=1}^\infty \in \mathbb{R}^\mathbb{N} \mid |x_i| \leq 1 \text{ for all } i \in \mathbb{N} \right\}$$

in the space of sequences $\mathbb{R}^\mathbb{N}$ equipped with the metric

$$d(x, y) = \sum_{i=1}^{\infty} \frac{|x_i - y_i|}{2^i}.$$

- (a) Is (X, d) closed?
- (b) Is (X, d) compact?